

MERN Hotel Management System

Software Requirements Specification (SRS)

LuxuryStay Hospitality — Version 1.0

Document Classification

Academic eProject Learning Environment — Full-Stack
Production-Grade Specification

Prepared For

LuxuryStay Hospitality, a multi-location high-end hotel chain requiring comprehensive operational transformation through modern software architecture.

Technology Stack

MongoDB · Express.js · React 18.x · Node.js v16+

API Scope

30+ production-ready RESTful API endpoints

Delivery Schedule

4-Phase Implementation — 7-Week Timeline

Document Version

SRS v1.0 — Initial Release

Executive Summary

This Software Requirements Specification defines the complete functional, architectural, and operational requirements for the **MERN Hotel Management System (HMS)** commissioned by LuxuryStay Hospitality. The document serves as the authoritative reference for all development, testing, and deployment activities across the 7-week delivery schedule.

Project Identity

- **Project Name:** MERN Hotel Management System (HMS)
- **Client:** LuxuryStay Hospitality — High-End Hotel Chain
- **Classification:** Academic eProject Learning Environment
- **Version:** 1.0 — Initial Specification Release

Technical Scope

- **Stack:** MongoDB, Express.js, React 18.x, Node.js v16+
- **API Coverage:** 30+ RESTful endpoints across 9 functional modules
- **Delivery:** 4-Phase implementation over a 7-week schedule
- **Architecture:** Full-stack, production-ready, cloud-deployable

Operational Transformation

The HMS consolidates all hotel operations — reservations, check-in/out, billing, housekeeping, and maintenance — into a single integrated platform, eliminating fragmented manual processes and data silos.

Strategic Decision Support

Real-time analytics and comprehensive reporting dashboards provide management with the data required for evidence-based strategic planning across single and multi-location operations.

Scalable by Design

A modular, microservice-aligned architecture built on MongoDB Atlas and Node.js ensures the system grows seamlessly from a single property to a full multi-location hotel chain deployment.

Problem Statement

LuxuryStay Hospitality, a high-end hotel chain with multiple locations, faces significant operational challenges that undermine both guest experience and long-term scalability. The following five critical pain points drive the need for this HMS initiative.

1	Fragmented Operations Manual processes across reservations, check-in/out, billing, and housekeeping create systemic inefficiencies. Staff operate across disconnected tools, resulting in data entry errors, duplicated effort, and delayed guest service.
2	Poor Guest Experience Absence of real-time room status visibility and personalized service delivery mechanisms prevents the hotel from meeting the expectations of its high-end clientele. No persistent guest preference data exists across visits.
3	Data Silos No unified platform exists for staff collaboration and management decision-making. Operations, billing, and housekeeping data are isolated, preventing cross-functional visibility and coordinated response.
4	Scalability Issues Current legacy systems cannot support the growing demands of multi-location operations. Horizontal scaling, load distribution, and centralized administration across properties are not architecturally feasible.
5	Compliance Gaps Inadequate security controls, data privacy enforcement, and audit trail mechanisms expose the organization to regulatory and reputational risk. Authentication, authorization, and data sovereignty mechanisms are absent.

Solution Mandate: Implement a modern, scalable, user-friendly Hotel Management System to streamline workflows, enhance guest experiences, and provide insightful data for strategic decision-making across all LuxuryStay properties.

Project Vision & Objectives

The HMS is designed around four primary operational goals and four learning objectives that together define the success criteria for Version 1.0. Each objective is traceable to specific functional requirements detailed in Chapters 1 through 3.

Primary Operational Goals

1 Operational Efficiency

Consolidate all hotel operations into a single, integrated platform eliminating manual handoffs and disconnected tools.

2 Guest Satisfaction

Deliver personalized, seamless guest experiences from initial booking through final checkout and loyalty reward issuance.

3 Data-Driven Decisions

Provide real-time analytics and reporting dashboards for operational and strategic planning at management level.

4 Scalability

Build a modular architecture capable of supporting growth from a single property to full multi-location chain operations.

Learning & Development Objectives

As an Academic eProject, this HMS demonstrates mastery across the following competencies:

→ Full-Stack MERN Mastery

End-to-end implementation using MongoDB, Express.js, React 18.x, and Node.js v16+ with industry-standard patterns.

→ Production-Grade Security

Authentication, role-based authorization, JWT token management, and OWASP-compliant security controls throughout.

→ Database & API Architecture

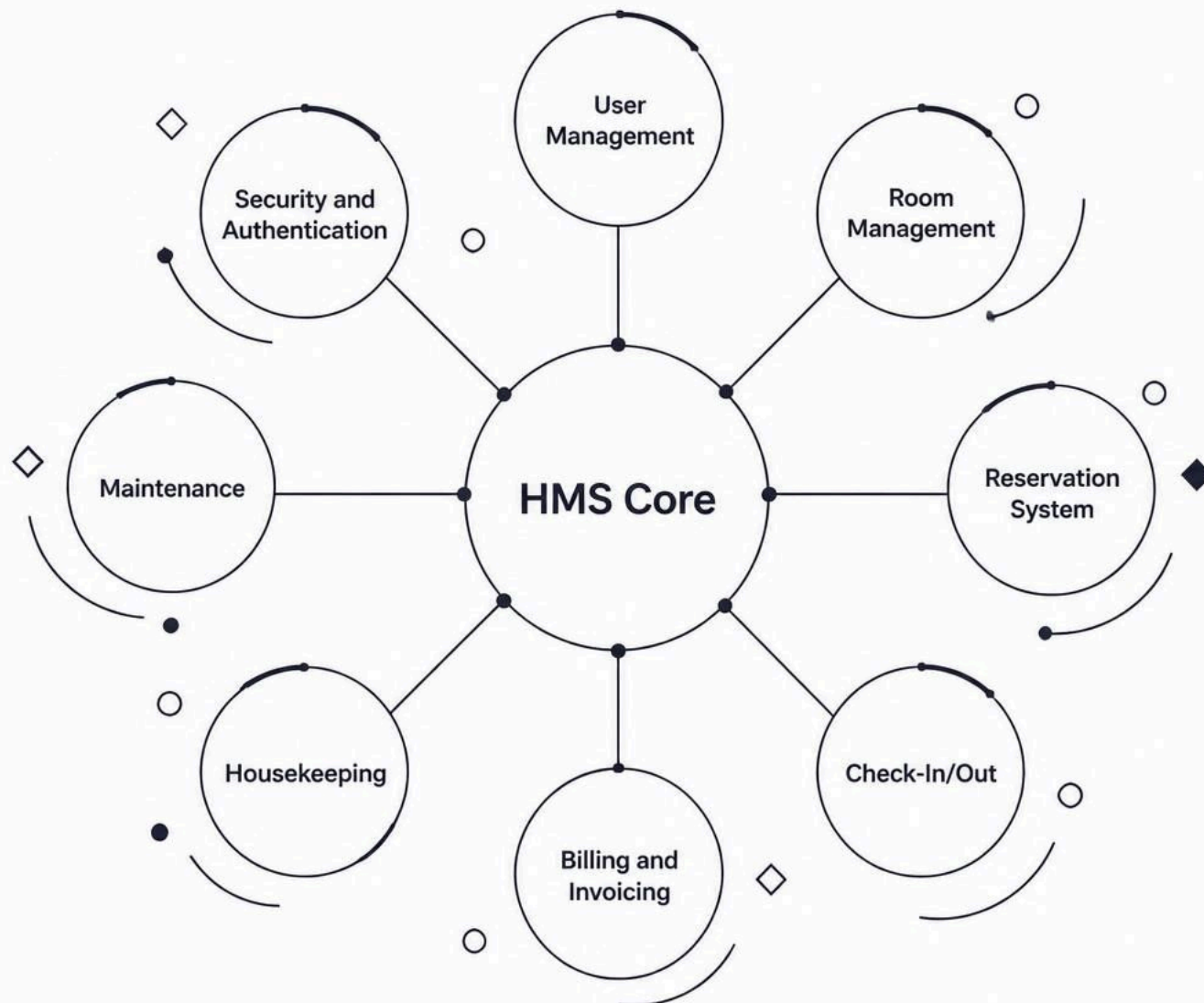
Scalable MongoDB schema design with optimistic locking, indexing strategy, and RESTful API endpoint architecture across 30+ routes.

→ Professional SRS Documentation

Deliver consultant-grade specification documentation meeting SRS standards for industry and academic evaluation.

Functional Requirements Overview

Chapter 1 defines all functional requirements across the nine operational modules of the HMS. Each module represents a bounded domain of system behavior with clearly specified actors, inputs, outputs, and business rules. The following module map provides a navigational overview of the complete functional specification.



Modules 1–5: Core Operations

- **Module 1:** User Management — Identity, roles, permissions
- **Module 2:** Room Management — Inventory, pricing, status
- **Module 3:** Reservation System — Booking lifecycle
- **Module 4:** Check-In / Check-Out — Guest arrival and departure
- **Module 5:** Billing & Invoicing — Charges, payments, receipts

Modules 6–9: Support Operations

- **Module 6:** Housekeeping — Cleaning tasks and scheduling
- **Module 7:** Maintenance — Issue tracking and resolution
- **Module 8:** Reporting & Analytics — Dashboards and KPIs
- **Module 9:** Security & Authentication — Access control layer

1.1 User Management Module

Establishes the foundational identity and access control layer with a role-based permission framework. All system interactions are governed by this module, ensuring that every actor operates within their authorized scope.

Admin Dashboard Capabilities

- Full administrative control over the complete user population
- Define roles, assign granular permissions, and set access levels per module
- Audit user activity logs and track all system changes with timestamped entries
- Unified view of all active and inactive accounts across properties

Staff Profile Management

- Complete CRUD operations for all staff accounts with change history
- Create, modify, suspend, or deactivate accounts with auditable change log
- Role assignment with enforcement of least-privilege access principle

Guest Profile Management

- Persistent guest records with personal information and contact details
- Store stated preferences: room type, floor level, amenity requirements
- Complete booking history and loyalty point tracking across all stays
- Enable personalized service delivery on all repeat visits

Role Hierarchy & Permission Matrix

Role	Authorized Access Scope
Administrator	Full system access, user management, all reporting modules, configuration
Manager	Operations oversight, billing overrides, staff scheduling, analytics
Receptionist	Reservations, check-in/out processing, guest profile management
Housekeeping	Room status updates, task queue management, completion reporting
Maintenance	Issue ticket management, status updates, work order processing
Accounts	Billing, invoice management, payment reconciliation, financial reporting

📌 The role hierarchy enforces strict least-privilege access. Escalation requests must be approved by an Administrator and are recorded in the audit log.

1.2 Room Management Module

The operational core of the HMS — maintains a live, accurate inventory of every room's attributes, pricing, and availability status. A finite state machine governs all room state transitions, ensuring data integrity under concurrent multi-user operations.

Room Inventory Tracking

Room Type	Status States	Pricing Basis
Standard	Available	Base rate × season factor
Deluxe	Occupied	Base rate × demand multiplier
Suite	Cleaning	Dynamic rate by occupancy %
Executive	Maintenance	Corporate contract override
Presidential	Out-of-Order	Event / bulk booking rate

Room attributes include floor number, capacity, amenity package, and base price. Dynamic pricing adjusts rates by room type, season, and real-time occupancy demand to maximize revenue per available room (RevPAR).

Room Booking Engine

- Real-time availability queries with conflict detection at query time
- Optimistic locking at database level prevents double-booking under concurrent load
- Date-range availability search with automatic room suggestions by preference
- Room assignment based on guest preference profile and occupancy patterns
- Manager override capability with mandatory audit trail entry
- Bulk block booking for corporate and event reservations

Status Automation

- Real-time status propagation to all connected staff interfaces
- Automated state transitions triggered by guest check-in and check-out events
- Maintenance request tracking with resolution status and elapsed time



Available

Occupied

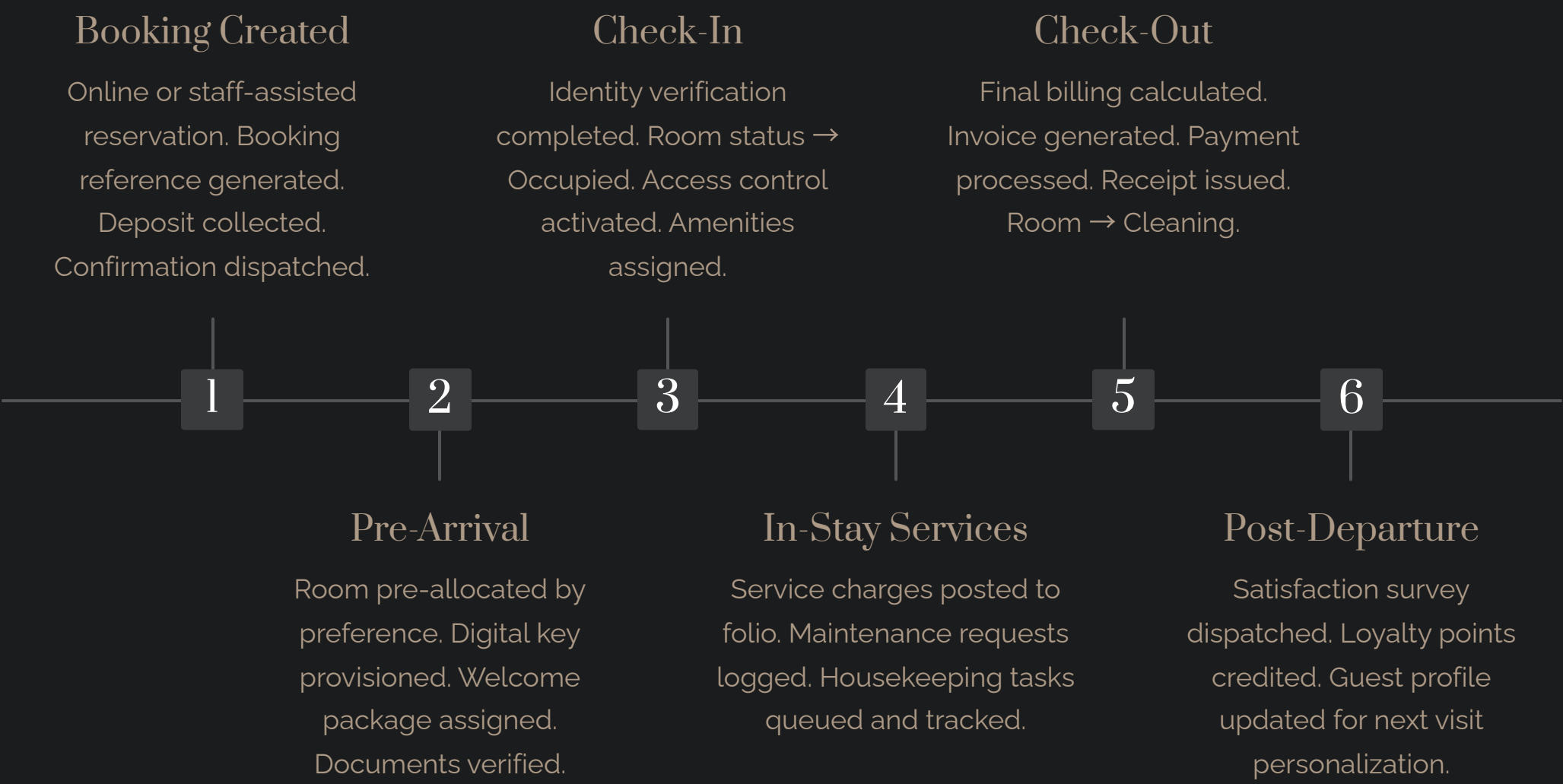
Cleaning

Maintenance

Out-of-Order

1.3 Reservation & Check-In/Out Module

Orchestrates the complete guest lifecycle from initial booking through departure, with automated billing triggers and real-time room status propagation at every stage of the process.



Reservation System — Key Requirements

- Online and staff-assisted booking with identical data validation rules
- Automated confirmation details and unique booking reference generation
- Flexible cancellation and modification policies with fee calculation
- Deposit and partial payment collection at time of booking

Check-Out Process — Key Requirements

- Final billing calculation inclusive of all posted service charges
- Payment processing with multi-method support and receipt generation
- Room status automated to "Cleaning" with housekeeping team notification
- Guest feedback collection via integrated satisfaction survey

1.4 Billing & Invoicing Module

Generates accurate, audit-compliant bills and detailed invoices with comprehensive charge breakdowns. Supports multi-method payment processing, tax compliance, and complete financial archival for regulatory and management purposes.

Billing System Requirements

- Automatic bill calculation based on room rate, room type, and duration of stay
- Additional service charge posting: food, laundry, room service, minibar, spa
- Tax calculation engine with compliance support for local and regional regulations
- Discount and promotional code application with eligibility validation
- Late checkout surcharge and early checkout credit adjustment processing

Payment Methods Supported

Card Credit & Debit	Cash Manual Entry
Bank Transfer Corporate	

Invoicing Requirements

- Detailed invoices with fully itemized charge breakdowns per line item
- Print-ready PDF and email delivery options for all guest-facing invoices
- Invoice history and archival with indexed search for audit and compliance purposes
- Refund processing with credit note generation and ledger reconciliation

Compliance & Audit Controls

- Every billing transaction stamped with actor ID, timestamp, and change delta
- Immutable invoice records — amendments generate new version with reference to original
- Manager-level override required for discounts exceeding configurable threshold
- Daily financial reconciliation report auto-generated for Accounts role review

📄 All billing records are retained for a minimum of 7 years in compliance with standard financial record-keeping requirements.

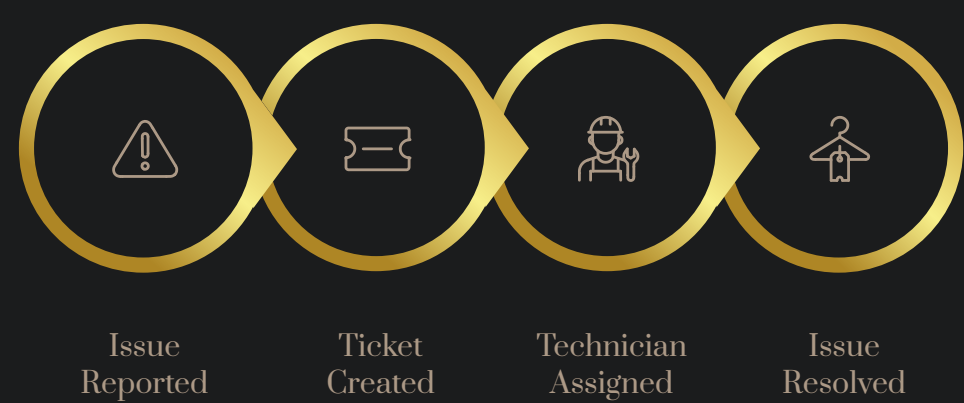
1.5 Housekeeping & Maintenance Module

Manages room cleaning schedules, maintenance requests, and issue resolution tracking. Integrates directly with Room Management to propagate real-time status updates and ensure guest-ready rooms are available without delay.

Housekeeping Management

- Real-time room status visibility for all housekeeping staff via role-specific dashboard
- Task queue with priority levels (urgent, standard, scheduled) and time estimates
- Mark tasks as completed with optional photo documentation for quality assurance
- Cleaning schedule optimization with intelligent staff assignment by floor and zone
- Escalation pathway for quality concerns requiring supervisor review

Maintenance Request Workflow



Maintenance Module Requirements

- Guests and staff can report maintenance issues through dedicated interface
- Automated ticket generation with priority classification: Critical, High, Normal, Low
- Track resolution status, assignee, elapsed time, and completion notes
- Maintenance history per room enables predictive maintenance scheduling
- Integration support for external maintenance vendor ticket systems if required

Implementation Roadmap Summary

